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SRN: PES2UG24CS311
Lab-II

Program 1:

```
#include <stdio.h>
```

```
int main(){
    printf("Enter the score of the student: \n");
    int score = 0;
    scanf("%d", &score);
    if(score >= 90)
        printf("Your Grade is: A\n");
    else if(score >= 80)
        printf("Your Grade is: B\n");
    else if(score >= 70)
        printf("Your Grade is: C\n");
    else if(score >= 60)
        printf("Your Grade is: D\n");
    else
        printf("Your Grade is: F\n");
}
```

Program 2:

```
#include <stdio.h>
```

```
int main(){
    printf("Enter the number of the month: \n");
    int month = 0;
    scanf("%d", &month);
    switch(month){
        case 1:
            printf("January: 31 days\n");
            break;
        case 2:
            printf("February: 28 days\n");
            break;
        case 3:
            printf("March: 31 days\n");
            break;
        case 4:
            printf("April: 30 days\n");
            break;
        case 5:
            printf("May: 31 days\n");
            break;
        case 6:
            printf("June: 30 days\n");
            break;
        case 7:
            printf("July: 31 days\n");
            break;
        case 8:
```

```

        printf("August: 31 days\n");
        break;
    case 9:
        printf("September: 30 days\n");
        break;
    case 10:
        printf("October: 31 days\n");
        break;
    case 11:
        printf("November: 30 days\n");
        break;
    case 12:
        printf("December: 31 days\n");
        break;
    default:
        printf("Invalid Input\n");
    }
}

```

Program 3:

```

#include <stdio.h>
int main(){
    printf("Enter the number: \n");
    int number = 0;
    scanf("%d", &number);
    if(number % 2 == 0)
        printf("Even Number\n");
    else
        printf("Odd Number\n");
}

```

Program 4:

```

#include <stdio.h>
int main(){
    printf("Enter the number of seconds: \n");
    int number = 0;
    scanf("%d", &number);
    if(number <= 59){
        if(number <= 30)
            printf("Ligth is: Red\n");
        else if(number <= 35)
            printf("Light is: Yellow\n");
        else
            printf("Light is: Green\n");
    }
    else
        printf("Invalid Input\n");
}

```

Program 5:

```

#include <stdio.h>
int main(){
    printf("Enter the temperature: \n");
    int number = 0;
    scanf("%d", &number);
}

```

```

        if(number >= 100)
            printf("It is Hot\n");
        else if(number >= 70)
            printf("It is warm\n");
        else if(number >= 45)
            printf("It is cool\n");
        else
            printf("It is cold\n");
        return 0;
    }

```

Program 6:

```

#include <stdio.h>
int main(){
    printf("Enter the Number: \n");
    int number = 0;
    scanf("%d", &number);
    if(number % 2 ==0 && number % 3==0)
        printf("It is divisble by both 2 and 3\n");
    else if(number % 2 ==0)
        printf("It is divisible by 2 only\n");
    else
        printf("It is divisble by 3 only\n");
    return 0;
}

```

Program 7:

```

#include <stdio.h>
int main(){
    printf("Enter two numbers seperated by spaces: \n");
    int x,y;
    scanf("%d %d", &x, &y);
    printf("Enter the operation: \n");
    char ch;
    getchar();
    scanf("%c", &ch);
    switch(ch){
        case '+':
            printf("Answer is: %d\n",x+y);
            break;
        case '-':
            printf("Answer is: %d\n",x-y);
            break;
        case '/':
            printf("Answer is: %.2f\n",(float)x/y);
            break;
        case '*':
            printf("Answer is: %d\n",x*y);
            break;
        default:
            printf("Invalid operator\n");
    }
    return 0;
}

```

```
}
```

Program 8:

```
#include <stdio.h>
```

```
int main(){
```

```
    printf("Enter weight in kg and height in meter seperated by spaces: \n");
```

```
    float w,h;
```

```
    scanf("%f %f", &w, &h);
```

```
    float bmi = w/(h*h);
```

```
    if (bmi >30)
```

```
        printf("You are Obese \n");
```

```
    else if(bmi >= 25)
```

```
        printf("You are overweight\n");
```

```
    else if(bmi >= 18.5)
```

```
        printf("Your Weight is Normal\n");
```

```
    else
```

```
        printf("You are underweight\n");
```

```
    return 0;
```

```
}
```

Lab - III
Srnr: PES2UG24CS311
Name: Neeraj R Ragi

1:

```
#include<stdio.h>
int main(){
    printf("Enter a number: \n");
    int num;
    scanf("%d", &num);
    int sum = 0, temp = num, x;
    int digit = 0;
    while(temp){
        digit++;
        temp /=10;
    }
    temp = num;
    while(num){
        x = num % 10;
        sum += pow(x, digit);
        num /= 10;
    }
    if(sum == temp)
        printf("It is armstrong number\n");
    else
        printf("It is not an armstrong\n");
    return 0;
}
```

2:

```
#include<stdio.h>
int main(){
    printf("Enter a number: \n");
    int num;
    scanf("%d", &num);
    int sum = 0, temp = num, x;
    temp = num;
    int digit =0;
    for(int i = 1; i <= num; i++){
        temp = i;
        while(temp){
            digit++;
            temp /=10;
        }

        digit--;
        temp = i;
        //printf("digit: %d\n", digit);
        //printf("temp: %d i: %d\n", temp, i);
        while(temp){
            sum += (temp % 10) * pow(10, digit);
            //printf("sum: %d\n", sum);
        }
    }
}
```

```

        temp /= 10;
        digit--;
    }
    if(sum == i)
        printf("%d is palindrome\n", i);
    digit = 0;
    sum = 0;
}
return 0;
}

```

3:

```

#include<stdio.h>
int main(){
    printf("Enter a number: \n");
    int num1, num2, n;
    scanf("%d", &n);
    int is_prime_1 = 1;
    int is_prime_2 = 1;
    int is_prime_3 = 1;
    int num3;
    for(int j = 2; j <= n; j++)
    {
        num1 = j;
        for(int l = 2; l <= n; l++)
        {
            num2 = l;
            is_prime_1 = 1;
            is_prime_2 = 1;
            is_prime_3 = 1;
            //printf("for (%d, %d):\n", num1, num2);
            for(int i = 2; i < (num1); i++){
                if (num1 % i == 0){
                    is_prime_1 = 0;
                    //printf("%d is not prime\n", num1);
                }
            }
            for(int i = 2; i < (num2); i++){
                if (num2 % i == 0){
                    is_prime_2 = 0;
                    //printf("%d is not prime\n", num1);
                }
            }
            if (is_prime_1 == 1 && is_prime_2 == 1){
                printf("The numbers (%d, %d) are both prime and:\n", num1,
num2);

                num3 = num2 - num1;
                if(num3 < 0)
                    num3 = -1 * num3;
                for(int i = 2; i < (num3); i++){
                    if (num3 % i == 0)
                        is_prime_3 = 0;

```

```

    }
    if(num3 == 1 || num3 == 0)
        is_prime_3 = 0;
    if (is_prime_3 == 1){
        printf("They are super primes\n");
    }
    else
        printf("They are Not super Primes\n");
    printf("\n\n");
}
}
}

return 0;
}

```

4:

```

#include<stdio.h>
int main(){
    printf("Enter 0 to stop the program: \n");
    int num = 0;
    int factor_sum;
    while(1){
        scanf("%d", &num);
        if(num == 0)
            break;
        factor_sum = 0;
        for(int i = 1; i <= (num/2); i++){
            if (num % i == 0)
                factor_sum += i;
        }
        if(factor_sum == num)
            printf("%d is a perfect number\n", num);
        else
            printf("%d is not a perfect number\n", num);
    }
    return 0;
}

```

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1.

```
#include<stdio.h>
int main(){
    printf("Enter The number of elements you want\n");
    int len = 0;
    scanf("%d", &len);
    int a[len];
    int out[len][2];
    int temp;
    int alraedy_present = 0;
    int ele = 0;

    printf("Enter Values: \n");
    for(int i = 0; i < len; i++){
        scanf("%d", &a[i]);
    }
    for(int i = 0; i < len; i++){
        temp = a[i];
        for(int j= 0; j < len;j++){
            if(temp == out[j][0]){
                alraedy_present = 1;
                out[j][1] += 1;
                break;
            }
        }
        if(!alraedy_present){
            out[ele][0] = temp;
            out[ele][1] = 1;
            ele++;
        }
        alraedy_present = 0;
    }
    for(int i = 0; i < ele; i++){
        printf("%d Occours: %d times\n", out[i][0], out[i][1]);
    }
    return 0;
}
```



```
neerajrrugi@student-OptiPlex-5090:~/Lab/ses-4/bin$ ./1.out
Enter The number of elements you want
8
Enter Values:
1
1
23
34
34
34
5
6
1 Occours: 2 times
23 Occours: 1 times
34 Occours: 3 times
5 Occours: 1 times
6 Occours: 1 times
```

2.

```
#include<stdio.h>
int main(){
    printf("Enter The number of elements you want\n");
    int len;
    scanf("%d", &len);
    int a[len];
    int out[len];
    int temp;
    int alraedy_present = 0;
    int ele = 0;

    printf("Enter Values: \n");
    for(int i = 0; i < len; i++){
```

```

        scanf("%d", &a[i]);
    }
    for(int i = 0; i < len; i++){
        temp = a[i];
        for(int j = 0; j < len; j++){
            if(temp == out[j]){
                already_present = 1;
                break;
            }
        }
        if(!already_present){
            out[ele] = temp;
            ele++;
        }
        already_present = 0;
    }
    printf("The array after removing duplicates is: \n");
    for(int i = 0; i < ele; i++){
        printf("%d\n", out[i]);
    }
    return 0;
}

```

```

neerajrrugi@student-OptiPlex-5090:~/Lab/ses-4/bin$ ./2.out
Enter The number of elements you want
8
Enter Values:
1
1
2
3
3
3
4
5
The array after removing duplicates is:
1
2
3
4
5

```

3.

```
#include<stdio.h>
int main(){
    printf("Enter the number of rows and columns you require: \n");
    int r,c;
    r = 0;
    c = 0;
    scanf("%d %d", &r, &c);
    printf("Enter the values for Array A row wise: \n");
    int a[r][c], b[r][c], x[r][c];
    for(int i = 0; i < r; i++){
        for(int j = 0; j < c; j++){
            scanf("%d", &a[i][j]);
        }
    }
    printf("Enter the values for Array B row wise: \n");
    for(int i = 0; i < r; i++){
        for(int j = 0; j < c; j++){
            scanf("%d", &b[i][j]);
        }
    }
    for(int i = 0; i < r; i++){
        for(int j = 0; j < c; j++){
            x[i][j] = a[i][j] + b[i][j];
        }
    }
    printf("The Matrix Sum is: \n");
    for(int i = 0; i < r; i++){
        for(int j = 0; j < c; j++){
            printf("%d ", x[i][j]);
        }
        printf("\n");
    }
}
```

```

neerajrrugi@student-OptiPlex-5090:~/Lab/ses-4/bin$ ./3.out
Enter the number of rows and columns you require:
2 2
Enter the values for Array A row wise:
1
2
3
4
Enter the values for Array B row wise:
6
7
8
9
The Matrix Sum is:
7 9
11 13

```

4.

```

#include<stdio.h>
int main(){
    printf("Enter the order of the matrix \n");
    int r;
    scanf("%d", &r);
    printf("Enter the values for Array A row wise: \n");
    int a[r][r];
    int p_dig =0, s_dig = 0;

    int x = 0, y = r-1;
    for(int i = 0; i < r; i++){
        for(int j = 0; j < r; j++){
            scanf("%d", &a[i][j]);
        }
    }
    for(int i = 0; i < r; i++){
        p_dig += a[i][i];
    }

    for(int i = 0; i < r; i++){
        s_dig += a[x][y];
        x++;
        y--;
    }

    printf("The Primary diagonal Sum is: %d\n",p_dig);
    printf("The Secondary diagonal Sum is: %d\n",s_dig);
}

```

```
neerajrrugi@student-OptiPlex-5090:~/Lab/ses-4/bin$ ./4.out
Enter the order of the matrix
2
Enter the values for Array A row wise:
2
1
3
5
The Primary diagonal Sum is: 7
The Secondary diagonal Sum is: 4
```

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1.

```
#include<stdio.h>
int main(){
    printf("Enter The number of elements you want\n");
    int len = 0;
    scanf("%d", &len);
    int a[len];
    int out[len][2];
    int temp;
    int alraedy_present = 0;
    int ele = 0;

    printf("Enter Values: \n");
    for(int i = 0; i < len; i++){
        scanf("%d", &a[i]);
    }
    for(int i = 0; i < len; i++){
        temp = a[i];
        for(int j= 0; j < len;j++){
            if(temp == out[j][0]){
                alraedy_present = 1;
                out[j][1] += 1;
                break;
            }
        }
        if(!alraedy_present){
            out[ele][0] = temp;
            out[ele][1] = 1;
            ele++;
        }
        alraedy_present = 0;
    }
    for(int i = 0; i < ele; i++){
        printf("%d Occours: %d times\n", out[i][0], out[i][1]);
    }
    return 0;
}
```

```
neerajrrugi@student-OptiPlex-5090:~/Lab/ses-4/bin$ ./1.out
Enter The number of elements you want
8
Enter Values:
1
1
23
34
34
34
5
6
1 Occours: 2 times
23 Occours: 1 times
34 Occours: 3 times
5 Occours: 1 times
6 Occours: 1 times
```

2.

```
#include<stdio.h>
int main(){
    printf("Enter The number of elements you want\n");
    int len;
    scanf("%d", &len);
    int a[len];
    int out[len];
    int temp;
    int alraedy_present = 0;
    int ele = 0;

    printf("Enter Values: \n");
    for(int i = 0; i < len; i++){
```

```

        scanf("%d", &a[i]);
    }
    for(int i = 0; i < len; i++){
        temp = a[i];
        for(int j = 0; j < len; j++){
            if(temp == out[j]){
                already_present = 1;
                break;
            }
        }
        if(!already_present){
            out[ele] = temp;
            ele++;
        }
        already_present = 0;
    }
    printf("The array after removing duplicates is: \n");
    for(int i = 0; i < ele; i++){
        printf("%d\n", out[i]);
    }
    return 0;
}

```

```

neerajrrugi@student-OptiPlex-5090:~/Lab/ses-4/bin$ ./2.out
Enter The number of elements you want
8
Enter Values:
1
1
2
3
3
3
4
5
The array after removing duplicates is:
1
2
3
4
5

```


3.

```
#include<stdio.h>
int main(){
    printf("Enter the number of rows and columns you require: \n");
    int r,c;
    r = 0;
    c = 0;
    scanf("%d %d", &r, &c);
    printf("Enter the values for Array A row wise: \n");
    int a[r][c], b[r][c], x[r][c];
    for(int i = 0; i < r; i++){
        for(int j = 0; j < c; j++){
            scanf("%d", &a[i][j]);
        }
    }
    printf("Enter the values for Array B row wise: \n");
    for(int i = 0; i < r; i++){
        for(int j = 0; j < c; j++){
            scanf("%d", &b[i][j]);
        }
    }
    for(int i = 0; i < r; i++){
        for(int j = 0; j < c; j++){
            x[i][j] = a[i][j] + b[i][j];
        }
    }
    printf("The Matrix Sum is: \n");
    for(int i = 0; i < r; i++){
        for(int j = 0; j < c; j++){
            printf("%d ", x[i][j]);
        }
        printf("\n");
    }
}
```

```

neerajrrugi@student-OptiPlex-5090:~/Lab/ses-4/bin$ ./3.out
Enter the number of rows and columns you require:
2 2
Enter the values for Array A row wise:
1
2
3
4
Enter the values for Array B row wise:
6
7
8
9
The Matrix Sum is:
7 9
11 13

```

4.

```

#include<stdio.h>
int main(){
    printf("Enter the order of the matrix \n");
    int r;
    scanf("%d", &r);
    printf("Enter the values for Array A row wise: \n");
    int a[r][r];
    int p_dig =0, s_dig = 0;

    int x = 0, y = r-1;
    for(int i = 0; i < r; i++){
        for(int j = 0; j < r; j++){
            scanf("%d", &a[i][j]);
        }
    }
    for(int i = 0; i < r; i++){
        p_dig += a[i][i];
    }

    for(int i = 0; i < r; i++){
        s_dig += a[x][y];
        x++;
        y--;
    }

    printf("The Primary diagonal Sum is: %d\n",p_dig);
    printf("The Secondary diagonal Sum is: %d\n",s_dig);
}

```

```
neerajrrugi@student-OptiPlex-5090:~/Lab/ses-4/bin$ ./4.out
Enter the order of the matrix
2
Enter the values for Array A row wise:
2
1
3
5
The Primary diagonal Sum is: 7
The Secondary diagonal Sum is: 4
```